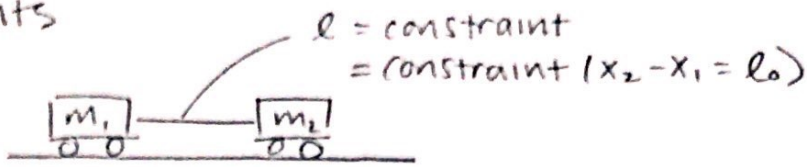


Recall constraints

ex)

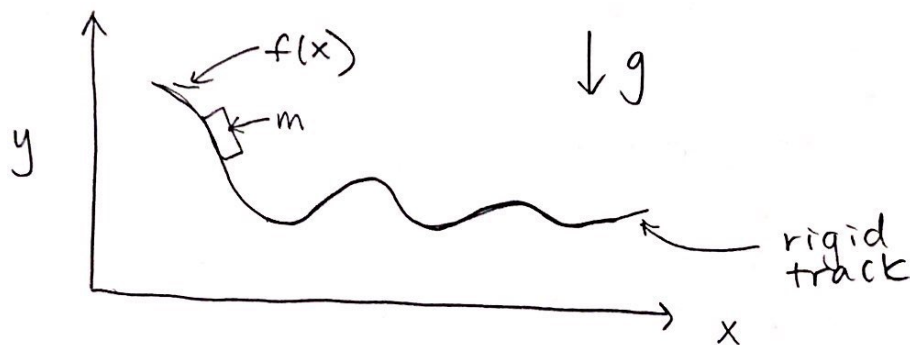


How to deal with constraints?

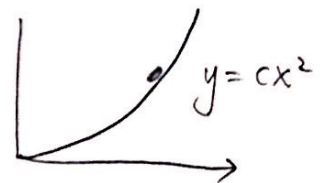
- 1) Minimal coordinates ("generalized" coordinates)
 - a) Newton-Euler (use appropriate eqns)
 - b) Lagrange
- 2) "Maximal" coordinates (w/ N-E)

ex) Roller coaster

$$y = f(x) = \text{given}$$



ex) $C = \text{given}$



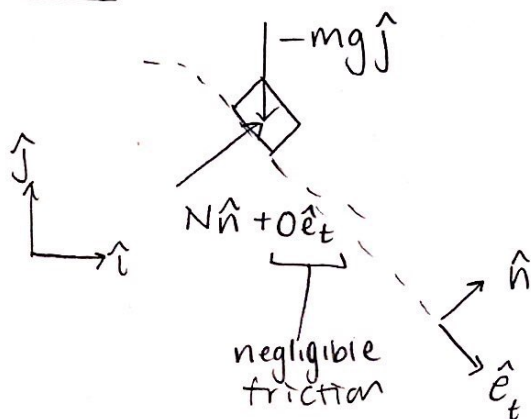
- can you find motion?
- given initial speed

How to find motion?

Assume:

- Negligible friction
= "frictionless"
- = "no" friction

FBD:



"bead on wire"

LMB:

$$\sum \vec{F}_i = m\vec{a}$$

$$-mg\hat{j} + N\hat{n} = \frac{d^2}{dt^2}(x\hat{i} + y\hat{j})m$$

$$= (\ddot{x}\hat{i} + \ddot{y}\hat{j})$$

Method 1a

Minimal coordinate x (1 DoF)

$$y = f(x)$$

$$\dot{y} = f'(x) \dot{x}$$

$$\ddot{y} = \underbrace{f''(x) \dot{x}^2 + f'(x) \ddot{x}}_h$$

$$\text{ex) } y = cx^2$$

$$\text{ex) } \dot{y} = 2cx \dot{x}$$

$$\text{ex) } \ddot{y} = \underbrace{2c \dot{x} + 2cx \ddot{x}}_h$$

N-E Minimal coordinates using x

$$\sum \vec{F} = m\vec{a}$$

$$\{-mg\hat{j} + N\hat{n} = m(\ddot{x}\hat{i} + g(x, \dot{x}, \ddot{x})\hat{j})\}$$

$$\vec{v} \cdot \{ \} \Rightarrow -mg\hat{j} \cdot \vec{v} = m(\ddot{x}\hat{i} + h\hat{j}) \cdot \vec{v}$$

$\vec{v} = \dot{x}\hat{i} + 2cx\dot{x}\hat{j}$

$\uparrow$ or $\hat{e}_t$

Trying to get rid of $N\hat{n}$

→ one equation in terms of $x, \dot{x}, \ddot{x}$

Solve for $a_x = \ddot{x}$

$$\Rightarrow \boxed{\begin{aligned} V_x &= \dot{x} \\ a_x &= \text{something}(x, \dot{x}, \text{parameters}) \\ &= \text{something}(x, \dot{x}, f, f') \end{aligned}}$$

Lagrange Equations

Minimal coordinates: x

$$E_p(x) = mgy = mgf(x) \stackrel{\text{ex)}}{=} mgcx^2$$

$$E_k(x, \dot{x}) = \frac{1}{2} mV^2$$

$$\hookrightarrow \vec{v} \cdot \vec{v}$$

$$\hookrightarrow \vec{v} = (\dot{x}\hat{i} + 2cx\dot{x}\hat{j})$$

$$\mathcal{L} = E_k - E_p = \mathcal{L}(x, \dot{x})$$

$$L.E : \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} = \cancel{Q_x} \rightarrow 0$$

→ same equations as N-E

Conservative System:

$$E_{TOT} = E_k + E_p \quad (\text{in terms of } x, \dot{x})$$

$$\frac{d}{dt} (E_{TOT}) = 0$$

cancel $\dot{x}$ (assume $\dot{x} \neq 0$)

can use soln to set up $x(t) = v$
eqn + integrate

same ODES

Method 4: DAE'S, the matrix thing

Maximal Coordinates

$$\star \vec{F} = m\vec{a} \quad 2 \text{ eqns for } \ddot{x}, \ddot{y}, N$$

add constraint eqn

$$\star \star \frac{d^2}{dt^2} (y = f(x))$$

⇒ 3rd 2nd order differential eqn

Solve $\star$ and $\star \star$ for $\ddot{x}, \ddot{y}, N$

Note, need consistent initial conditions